

4.1.4 Waves

Water waves and sound waves are used here to distinguish between transverse and longitudinal waves, which transfer energy and information without transferring matter. This leads to the study of

the continuous spectrum of electromagnetic waves. The hazards associated with some electromagnetic waves feature in [Radiation and risk](#) (page 55).

There are two required practicals: one studying waves in a ripple tank and a metal rod, and the other looking at infrared radiation from different surfaces. Knowledge of properties of parts of the electromagnetic spectrum is needed to explain the greenhouse effect (see [The greenhouse effect](#) (page 69)).

4.1.4.1 Transverse and longitudinal waves

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Describe the difference between transverse and longitudinal waves. Describe how ripples on water surfaces are examples of transverse waves whilst sound waves in air are longitudinal waves, and how the speed of each may be measured. Describe evidence that in both cases it is the wave and not the water or air itself that travels.	In a transverse wave the oscillations are perpendicular to the direction of energy transfer. The ripples on a water surface are an example of a transverse wave. In a longitudinal wave the oscillations are parallel to the direction of energy transfer. Longitudinal waves show areas of compression and rarefaction. Sound waves travelling through air are longitudinal.	WS 2.3 Describe one method to measure the speed of sound waves in air. WS 2.2, 2.3 Describe one method to measure the speed of ripples on a water surface. WS 3.5 Interpret given data from experiments to measure the speed of sound or water waves.

Required practical activity 5: make observations to identify the suitability of apparatus to measure the frequency, wavelength and speed of waves in a ripple tank and waves in a solid and take appropriate measurements.

AT skills covered by this practical activity: physics AT 4.

This practical activity also provides opportunities to develop WS and MS. Details of all skills are given in [Key opportunities for skills development](#) (page 165).

4.1.4.2 A wave equation

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Describe wave motion in terms of amplitude, wavelength, frequency, and period; define wavelength and frequency and describe and apply the relationship between these and the wave velocity.	Waves are described by their amplitude, wavelength, frequency and period. The amplitude of a wave is the maximum displacement of a point on a wave away from its undisturbed position. The wavelength of a wave is the distance from a point on one wave to the equivalent point on the adjacent wave. The frequency of a wave is the number of waves passing a point each second. The wave speed is the speed at which the energy is transferred (or the wave moves) through the medium. All waves obey the wave equation: $\text{wave speed} = \text{frequency} \times \text{wavelength}$ $[v = f \lambda]$ wave speed, v, in metres per second, m/s frequency, f, in hertz, Hz wavelength, λ, in metres, m Students should be able to apply the relationship: $\text{period} = \frac{1}{\text{frequency}}$ $\left[T = \frac{1}{f} \right]$ period, T, in seconds, s frequency, f, in hertz, Hz	WS 4.6, MS 1b, 2a Calculate with numbers written in standard form and give answers to an appropriate number of significant figures. MS 1c, 3b, 3c Recall and apply the wave equation. MS1a, 1c, 3b, 3c Apply the equation for relationship between period and frequency, which is given on the Physics equations sheet. WS 3.3 Carry out and represent mathematical and statistical analysis.

4.1.4.3 Electromagnetic waves

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Recall that electromagnetic waves are transverse, are transmitted through space where all have the same velocity, and explain, with examples, that they transfer energy from source to absorber. Recall that light is an electromagnetic wave. Describe the main groupings of the spectrum – radio, microwave, infrared, visible (red to violet), ultraviolet, X-rays and gamma rays, that these range from long to short wavelengths and from low to high frequencies, and that our eyes can only detect a limited range. Give examples of some practical uses of electromagnetic waves in the radio, microwave, infrared, visible, ultraviolet, X-ray and gamma ray regions.	Electromagnetic waves form a continuous spectrum. Examples of uses of electromagnetic waves include: • radio waves – television, radio and radio telescopes • microwaves – satellite communications, cooking food • infrared – electrical heaters, cooking food, infrared cameras • visible light – fibre optic communications • ultraviolet – fluorescent lamps, sun tanning • X-rays – medical imaging and treatments • gamma rays – sterilising surgical instruments, treatment of cancer.	WS 1.2 Show that the uses of electromagnetic waves illustrate the transfer of energy from source to absorber. MS 1a, 1c, 3c Recall and apply the relationship between frequency and wavelength across the electromagnetic spectrum.

Required practical activity 6: investigate how the amount of infrared radiation absorbed or radiated by a surface depends on the nature of that surface.

AT skills covered by this practical activity: physics AT 1 and 4.

This practical activity also provides opportunities to develop WS and MS. Details of all skills are given in [Key opportunities for skills development](#) (page 166).

4.1.4.4 Radio waves (HT only)

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Recall that radio waves can be produced by, or can themselves induce, oscillations in electrical circuits.	When radio waves are absorbed they may create an alternating current with the same frequency as the radio wave itself, so radio waves can themselves induce oscillations in an electrical circuit.	

4.1.4.5 Reflection and refraction of electromagnetic waves (HT only)

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Recall that different substances may refract, or reflect these waves; explain how some effects are related to differences in the velocity of the waves in different substances.	Shiny surfaces act as mirrors when they reflect waves. Rough surfaces scatter waves in all directions. Electromagnetic waves change speed when they travel between different substances such as from air to glass or water. As a result they change direction. This is refraction.	WS 1.2 Construct ray diagrams to illustrate the refraction of a wave at the boundary between two different media. Use wavefront diagrams to explain refraction in terms of the change of wave speed.